

09/14/2026: Factorial

CSCI 246: Discrete Structures

Textbook reference: Sec. 9, Scheinerman

Graded Quiz Pickup

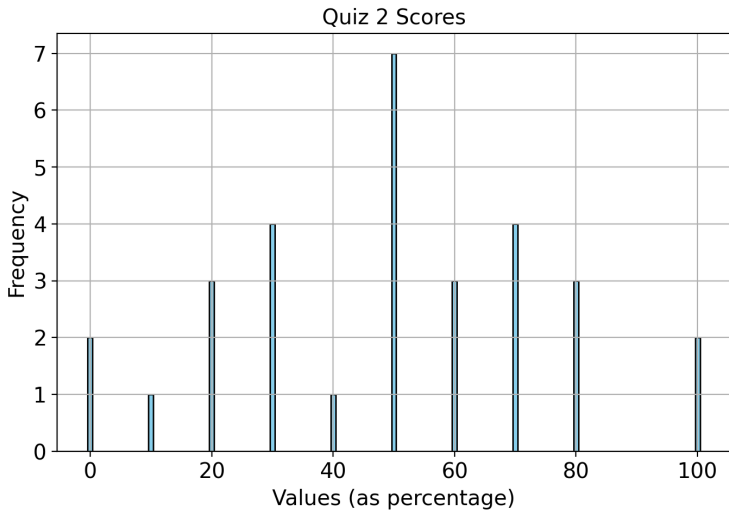
Quizzes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Today's Agenda

- Quiz feedback/logistics (≈ 10 mins)
- Review group exercises (≈ 10 mins)
- Mini-lecture (≈ 15 mins)
- Group exercises ≈ 15 mins)

Quiz feedback/logistics

Weekly Quiz #2: Scores



Median: 2.5/5 (50%).

Proofs: A second pass

I've added an extra day before the midterm to review proofs, with an option to retake quiz #2.

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2. (New!) **Free** daily tutoring is provided by SmartyCats. You can make an appointment Mon-Fri 3:30-5:30. See the updated [syllabus](#) .

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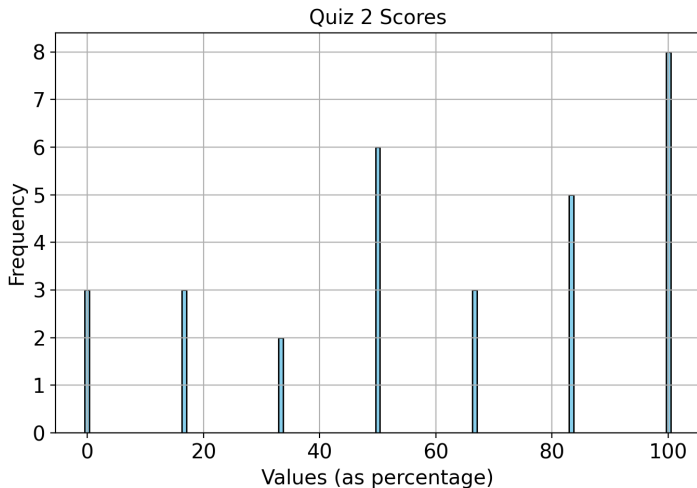
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3. (New!) Madhav (TA) is available to meet Friday mornings by appointment.

Weekly Quiz #2: Scores - Updated

I discarded the first question since I didn't provide the definition for odd.



Median: $2/3(66)\%$.

Quiz Logistics

- Going forward, if you finish the quiz early, you will need to be silent until everyone else is finished.

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Review of lists

Multiplication Principle

When choosing 1 burger, 1 side serve and 1 drink ...
how many combinations are possible?

burgers



- beef
- chicken
- fish
- vegetarian

4 possibilities

side serves



- fries
- salad

2 possibilities

drinks



- juice
- coffee
- cola
- water

4 possibilities

Solution. There are $4 \times 2 \times 4 = 32$ possibilities.

Multiplication principle

Multiplication Principle (Theorem 8.2 from Scheinerman)

Consider the two-element lists for which there are n choices for the first element, and for each choice of the first element there are m choices for the second element. Then the number of such lists is nm .

Remark. We can apply the multiplication principle *iteratively*. So, for example, if we want to form three-element lists for which there are n, m , and k choices for the 1st, 2nd, and 3rd elements respectively, then there are nmk such lists.

Multiplication principle with constant pool size

Theorem 8.6 from Scheinerman

The number of lists of length k whose elements are chosen from a pool of n elements

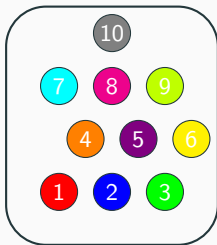
$$= \begin{cases} n^k, & \text{if repetitions are permitted} \\ (n)_k, & \text{if repetitions are forbidden} \end{cases}$$

Notation.

$$(n)_k := \underbrace{n \times (n-1) \times \cdots \times (n-k+1)}_{k \text{ terms}}$$

Multiplication principle with constant pool size

Example: Lottery Suppose we draw $k = 6$ balls from the jar of $n = 10$ balls and record the sequence in which they appear. How many possible sequences are there?



Solution to Lottery Problem. The number of possible sequences if we draw $k = 6$ balls from the jar of $n = 10$ balls and record the sequence in which they appear is:

$$= \begin{cases} 10^6, & \text{if repetitions are permitted} \\ (10)_6 = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5, & \text{if repetitions are forbidden} \end{cases}$$

Factorial

Poll. In how many different orders could these runners in the 2022 NYC marathon runners end up finishing the race?



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Solution. 5!

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Solution. 5! What does that mean?

Explanation.

Poll. In how many different orders could these runners in the 2022 NYC marathon runners end up finishing the race?



Solution. $5! = 5 \times 4 \times 3 \times 2 \times 1$.

Explanation. Why is this correct?

Poll. In how many different orders could these runners in the 2022 NYC marathon runners end up finishing the race?



Solution. $5! = 5 \times 4 \times 3 \times 2 \times 1$.

Explanation. Why is this correct? Remember the **multiplication principle**. There are 5 choices for 1st place. For each such choice, there are 4 choices for 2nd place. Etc.

Poll. In how many different orders could these runners in the 2022 NYC marathon runners end up finishing the race?



Solution. $5! = 5 \times 4 \times 3 \times 2 \times 1$.

Explanation. Why is this correct? Remember the **multiplication principle**. There are 5 choices for 1st place. For each such choice, there are 4 choices for 2nd place. Etc. This is like nested for loops!

Factorial

Definition

n **factorial**, written $n!$, is the number of length- n lists that can be formed from a pool of n objects in which repetition is forbidden.

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Technical Note

The definition implies that $n!$ is only defined when n is a non-negative integer ($0, 1, 2, 3, \dots$).

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Technical Note

The definition implies that $n!$ is only defined when n is a non-negative integer (0, 1, 2, 3, ...).

Interpretation

We can think of n -factorial as the number of way to **shuffle** n objects.

For example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ is the number of ways we can order the NYC Marathon runners Sharon Lokedi, Caroline Rotich, Lonah Salpeter, Edna Kiplagat, and Aliphine Tuliamuk.

Falling Factorial

Recall. Last time we defined the **falling factorial**

$$(n)_k := \underbrace{n \times (n-1) \times (n-2) \times \cdots \times (n-k+1)}_{k \text{ terms}},$$

read as *n falling k*, as (How to describe this this out loud?)

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Poll. How is the falling factorial related to the factorial?

Solution. Note that

$$(n)_1 = n$$

$$(n)_2 = n \times (n-1)$$

$$(n)_3 = n \times (n-1) \times (n-2)$$

$$\vdots = \vdots$$

$$(n)_n = n \times (n-1) \times (n-2) \cdots \times 1$$

Hence $n! = (n)_n$, and so the factorial is a special case of the falling factorial.

Factorial: Special case of falling factorial

Theorem 8.6 from Scheinerman

The number of lists of length k whose elements are chosen from a pool of n elements

$$= \begin{cases} n^k, & \text{if repetitions are permitted} \\ (n)_k & \text{if repetitions are forbidden} \end{cases}$$

Definition (from Scheinerman Sec. 9)

The number of lists of length n whose elements are chosen from a pool of n elements if repetitions are forbidden is called n **factorial**, written $n!$.

Factorial: Special case of falling factorial

Theorem 8.6 from Scheinerman

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Definition (from Scheinerman Sec. 9)

The number of lists of length n whose elements are chosen from a pool of n elements if repetitions are forbidden is called n **factorial**, written $n!$.

Note. The factorial is a special case of the falling factorial where **what?**

Factorial: Special case of falling factorial

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Definition (from Scheinerman Sec. 9)

The number of lists of length n whose elements are chosen from a pool of n elements if repetitions are forbidden is called n **factorial**, written $n!$.

Note. The factorial is a special case of the falling factorial where the length of the list, k , equals the length of the pool of elements we're selecting from, n .

Permutations and combinations

Suppose you are counting the number of ways to do something...

Permutations are what you use when the order matters. For instance, we may want to calculate the number of possible orderings for the 5 marathon runners. For this, we can use factorials (or more generally, falling factorials). (How so?)



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Combinations are what you use when the order doesn't matter. For instance, if you have 10 different pieces of clothing you want to take on a trip, but you can only fit 7 of them in your suitcase, it matters which 7 you pick, but it doesn't matter what order you put them in the suitcase.



Permutations and combinations

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Relation to textbook. Right now we're working with permutations. (Note: we're working with lists, and lists are **ordered!**) We'll get to combinations later, in Sec. 17 (Binomial Coefficients).

Applications in computer science: Some examples

Graph algorithms. Permutations are used when checking all possible paths. Example: the **Traveling Salesman Problem**, where a salesman must visit each city from a set (e.g. each U.S. capitol) exactly once and return to the starting city, finding the cheapest possible route to minimize travel cost. One approach checks all $n!$ possible orderings of cities to find the cheapest tour.



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Assigning IP addresses in a subnet. Suppose a network has n available IP addresses and you want to assign k devices distinct addresses. The number of ways to assign the addresses is $(n)_k$.

IP Address

192.168.1.1

Zero factorial

Poll. We define $0! = 1$. Why?

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Solution. We give two arguments below

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Solution. We give two arguments below

1. **By definition.** $0!$ gives the number of lists of length 0 whose elements are chosen from a pool of 0 elements in which repetition is forbidden.

Zero factorial

Poll. We define $0! = 1$. Why?

Solution. We give two arguments below

1. **By definition.** $0!$ gives the number of lists of length 0 whose elements are chosen from a pool of 0 elements in which repetition is forbidden. There is only one such list: $()$, the empty list.
2. **Multiplicative identity.**

Zero factorial

Poll. We define $0! = 1$. Why?

Solution. We give two arguments below

1. **By definition.** $0!$ gives the number of lists of length 0 whose elements are chosen from a pool of 0 elements in which repetition is forbidden. There is only one such list: $()$, the empty list.
2. **Multiplicative identity.** We would like $0!$ to equal the multiplicative identity (that is, 1).

Reference.

- *1 is the multiplicative identity.* Multiplying some number x by 1 leaves it unchanged.
- *0 is the additive identity.* Adding 0 to some number x leaves it unchanged.

Zero factorial

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Solution. We give two arguments below

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2. **Multiplicative identity.** We would like $0!$ to equal the multiplicative identity (that is, 1). For example, consider the identity

$$n! = n \times (n - 1)!$$

In the case where $n = 1$, this becomes

$$1! = 1 \times 0!$$

which holds only if $0! = 1$.

Product notation

We can write the factorial as

$$n! = \prod_{k=1}^n k,$$

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Remark. Products of the form $\prod_{k=1}^0 k$ are called empty products. These evaluate to 1.

Random group assignments

Ashley, Isabelle: 8

Bailey, Tanna: 9

Blaisdell, Olyver: 10

Brown, Ashton: 4

Brown, Jonathan: 12

Cardenas, Dominic: 3

Cutler, George: 13

Fabik, Roland: 12

Fellows, Audrey: 9

Fitzgerald, Ryan: 7

Fuller, Ryan: 1

Griffen, Clara: 2

Grutsch, Tanner: 2

Hager, Nathan: 6

Horner, Jayden: 6

Johnson, Jaret: 5

Kintzel, Samantha: 6

Lameres, Alexis: 13

Lee, Holden: 7

Mayala, Reece: 7

McLean, Connor: 4

Mcbride, Daniel: 3

Miller, Maverick: 8

Moss, Emily: 5

Mousseau, Matthew: 12

Neuman, Evan: 11

Putnam, John: 8

Ramos, Vydel: 13

Rathbun, Jedidiah: 1

Roberts, Matthew: 11

Roller, Nathan: 2

Ryan, Otis: 1

Shevtsov, Timothy: 9

Sinclair, Aidan: 4

Smith, Charles: 10

Stensland, Emma: 3

Thiessen, Logan: 10

Thompson, Jaiden: 5

Vath, Ethan: 11

Group exercises

1. Your friend Sandy has six different books of classical literature, eight different *romantasy* (i.e. romance fantasy) books, and five different self-help books.

- a) In how many different ways can Sandy's books be arranged on a bookshelf?
- b) In how many different ways can Sandy's books be arranged on the bookshelf if all books of the same category are grouped together?



2. Evaluate $\frac{100!}{98!}$ without calculating 100! or 98!.

3. Calculate the following products:

- a) $\prod_{k=1}^n \frac{k+1}{k}$. (Reduce this product.)
- b) $\prod_{k=1}^n (2k - 1)$. (Give an English description.)

Solution to group exercise #1

Problem. Your friend Sandy has six different books of classical literature, eight different *romantasy* (i.e. romance fantasy) books, and five different self-help books.

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Solution.

- a) By the definition of factorial, there are $(6 + 8 + 5)! = 19!$ different ways to arrange the books.

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- b) By the definition of factorial, there are $6!$ ways to arrange the classical literature books, $8!$ ways to arrange the *romantasy* books, and $5!$ ways to arrange the self-help books. So by the multiplication principle, there are $6!8!5!$ ways to arrange the books, if we assume a particular ordering of the categories (e.g. classical $\rightarrow$ *romantasy* $\rightarrow$ self-help). (*Are we done?*)

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Solution to group exercise #2

Problem. Evaluate $\frac{100!}{98!}$ without calculating $100!$ or $98!$.

Solution.

$$\frac{100!}{98!} = \frac{100 \cdot 99 \cdot 98!}{98!} = \frac{100 \cdot 99 \cdot \cancel{98!}}{\cancel{98!}} = 100 \cdot 99 = 9900.$$

Solution to group exercise #3

Problem. Calculate the following products:

- a) $\prod_{k=1}^n \frac{k+1}{k}$. (Reduce this product to a simpler expression.)
b) $\prod_{k=1}^n (2k-1)$. (This product has a simple English-language description.
What is it?)

Solution.

a)

$$\begin{aligned}\prod_{k=1}^n \frac{k+1}{k} &= \frac{2}{1} \cdot \frac{3}{2} \cdots \frac{n}{n-1} \cdot \frac{n+1}{n} \\ &= \frac{\cancel{2}}{1} \cdot \frac{\cancel{3}}{\cancel{2}} \cdots \frac{\cancel{n}}{\cancel{n-1}} \cdot \frac{n+1}{\cancel{n}} \\ &= n+1\end{aligned}$$

Solution to group exercise #3

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b)

$$\begin{aligned}\prod_{k=1}^n (2k-1) &= (2 \cdot 1 - 1) \cdot (2 \cdot 2 - 1) \cdot (2 \cdot 3 - 1) \cdots (2 \cdot n - 1) \\ &= 1 \cdot 3 \cdot 5 \cdots 2n-1\end{aligned}$$

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So what's the interpretation?

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Solution.

a)

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This is the product of the first n odd numbers. (The k -th factor is the k -th odd number.)